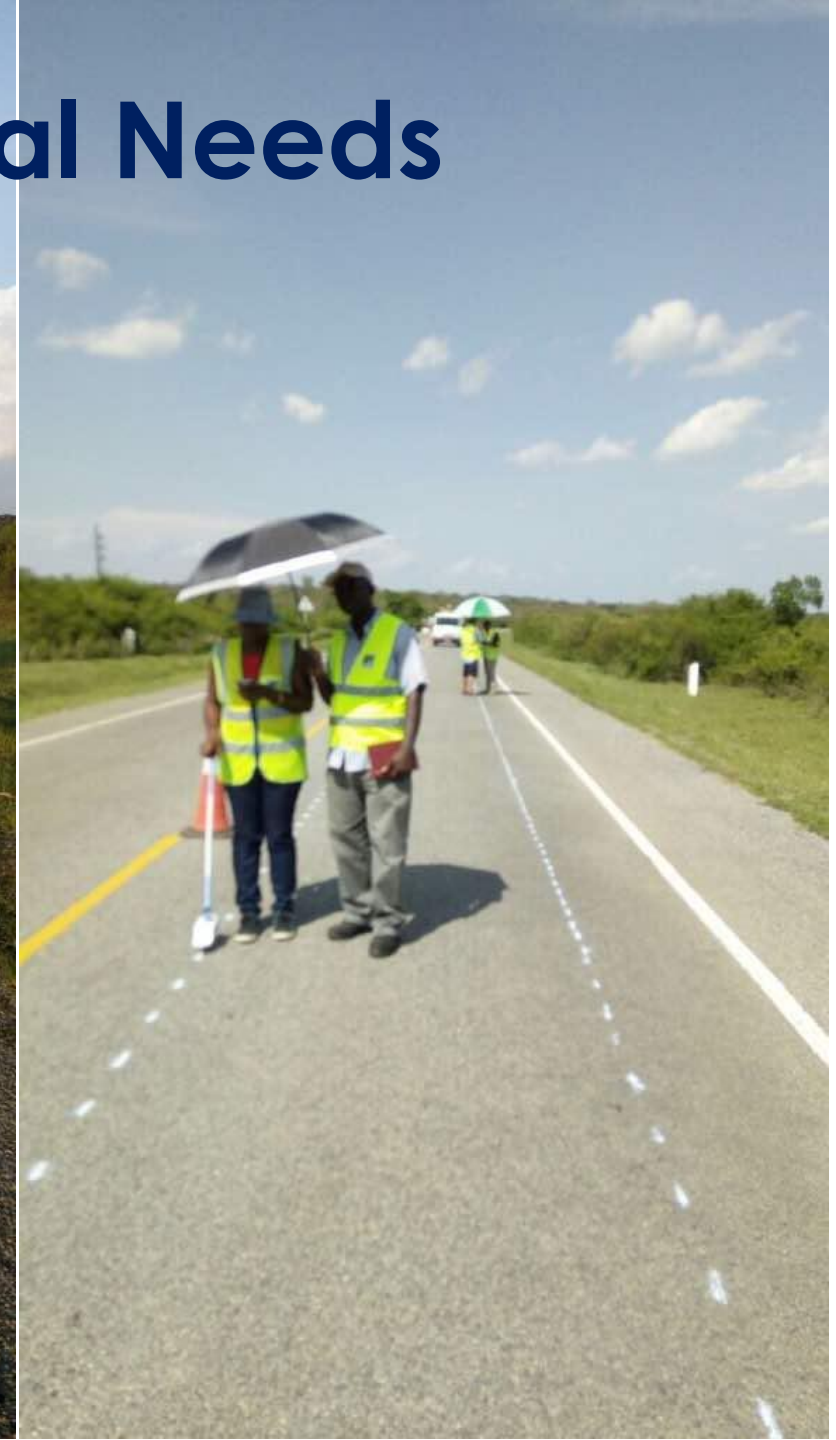
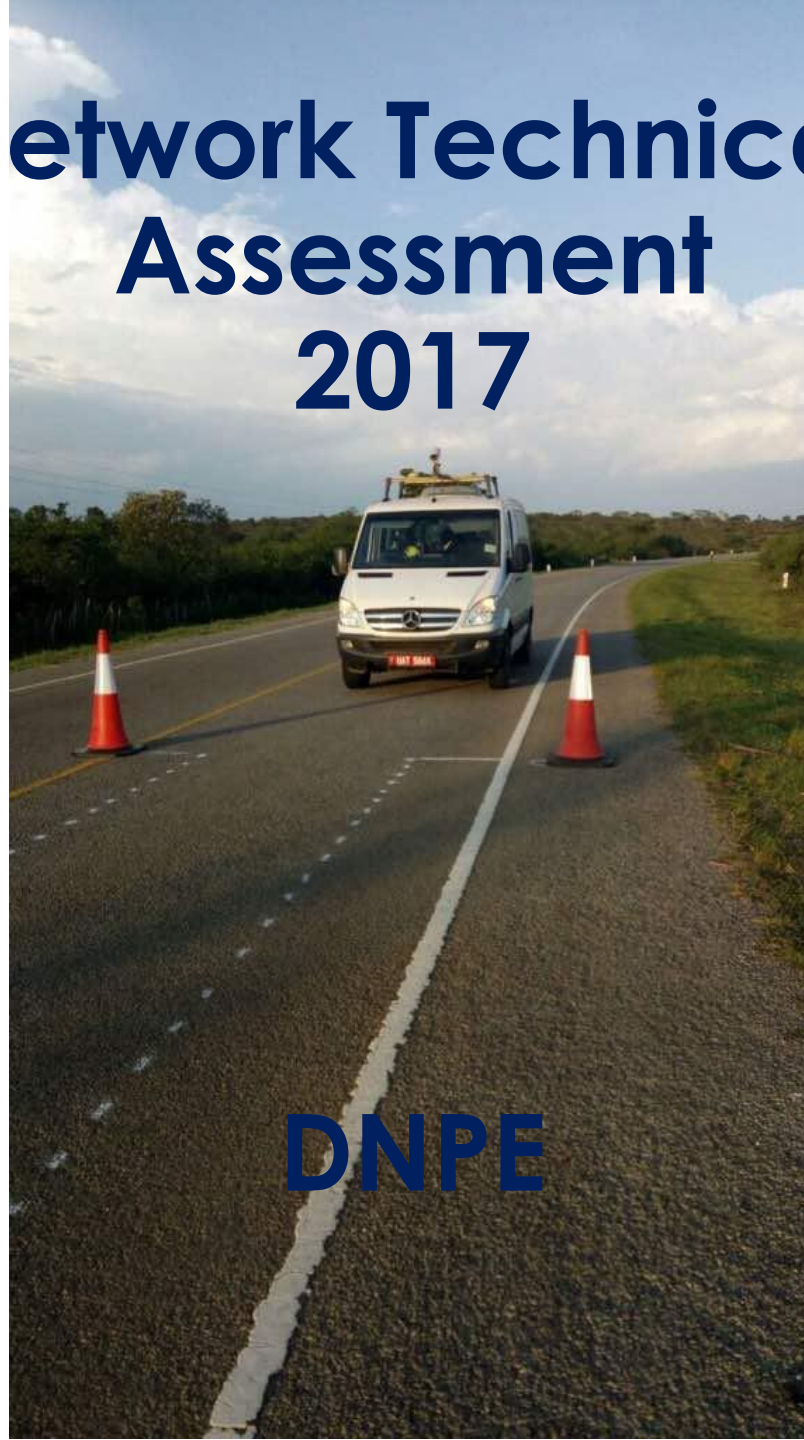


Roads Network Technical Needs Assessment 2017



DNPE



MANDATE

- URF Act 2006 (Clause 23) requires all agencies to prepare AMP's based on data collected or RMS
- Network Planning core function is to collect road condition data to support development and maintenance programming of the network.



Main Objective

- **To prepare a network roads investment plan and assist UNRA determine the technical needs, understand the risks and attain its set goals.**
- *The goals are; reduce transport costs, travel time and improve safety.*
- *The aim is to promote economic and social development and facilitate international and transit trade thereby boosting national competitiveness*



Specific Objectives

- Establish the current Inventory, condition and traffic usage of the network
- Show the impact of the various funding levels on the future road network condition, traffic, maintenance backlog, road user costs, asset value; taking into account current committed projects;
- Prepare a five year optimised Preventive Maintenance and Rehabilitation Plan for the current funding level.

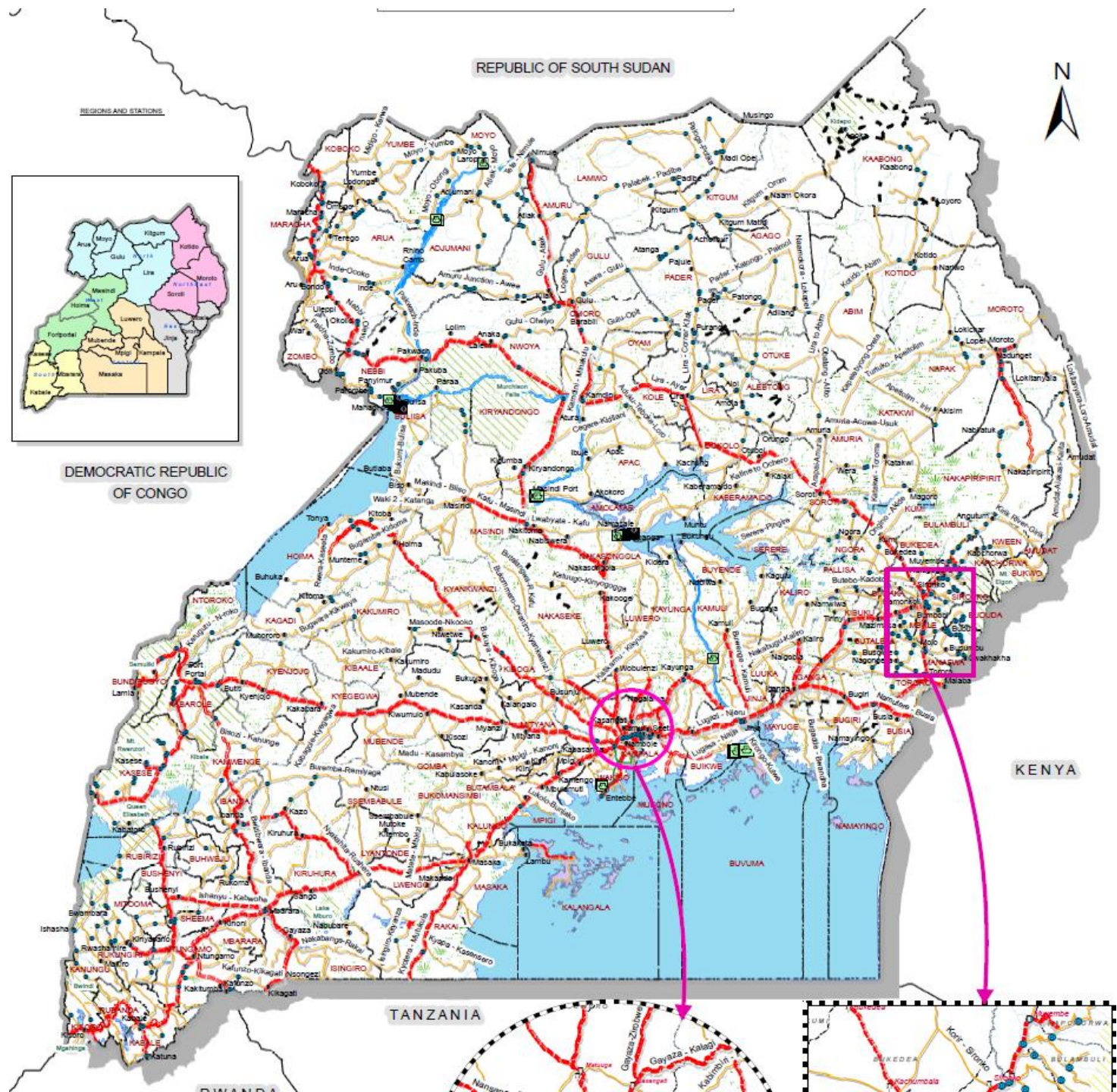


NATIONAL ROAD NETWORK (Uganda)

Road Type	Surface type			Total Network	Percentage of Ownership
	Tarmac	Murram	Dirt / Earth		
National UNRA	4,501(79%)	13,596	3,480.50	21,577.84	13.60%
District	645 (11%)	26,418.47	13,918.61	40,982.62	25.80%
Municipality	437 (8%)	3,206.22	2,632.40	6,275.32	4.00%
Town Council	85 (2%)	7,212.05	4,247.53	11,544.79	7.30%
Community	7 (0.1%)	21,402.43	57,016.12	78,425.60	49.40%
Total	5,675.70	71,835.40	81,295.10	158,806.17	100.0%
Percentage	3.60%	45.20%	51.20%	100.00%	



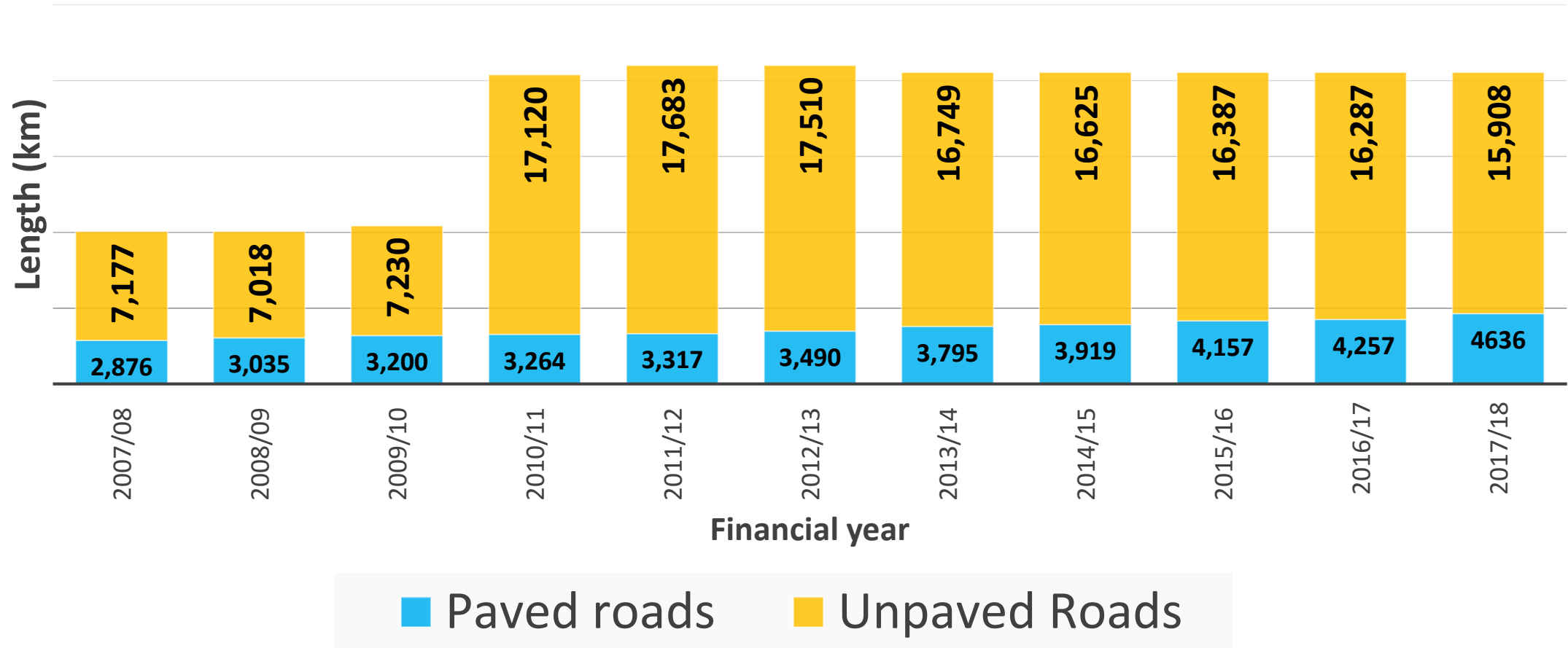
NATIONAL ROAD NETWORK COVERAGE





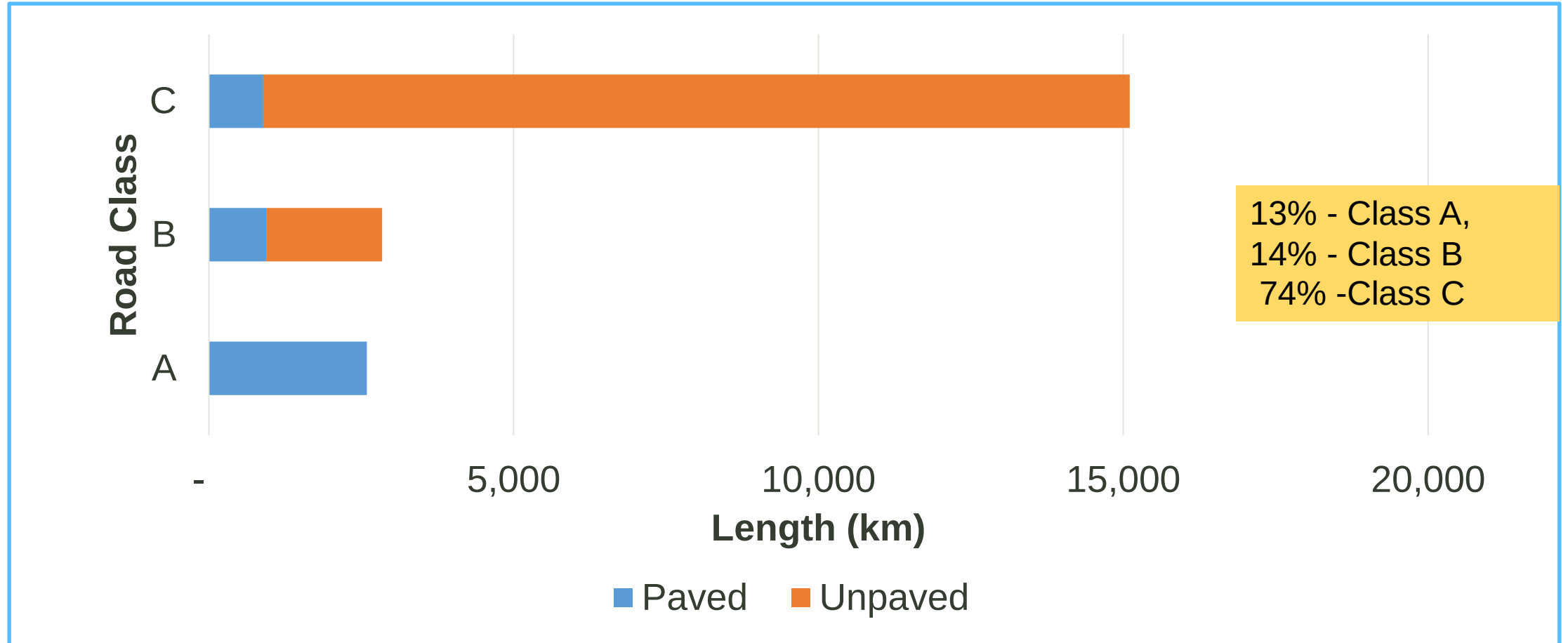
NATIONAL ROAD NETWORK GROWTH

Length of the UNRA road network





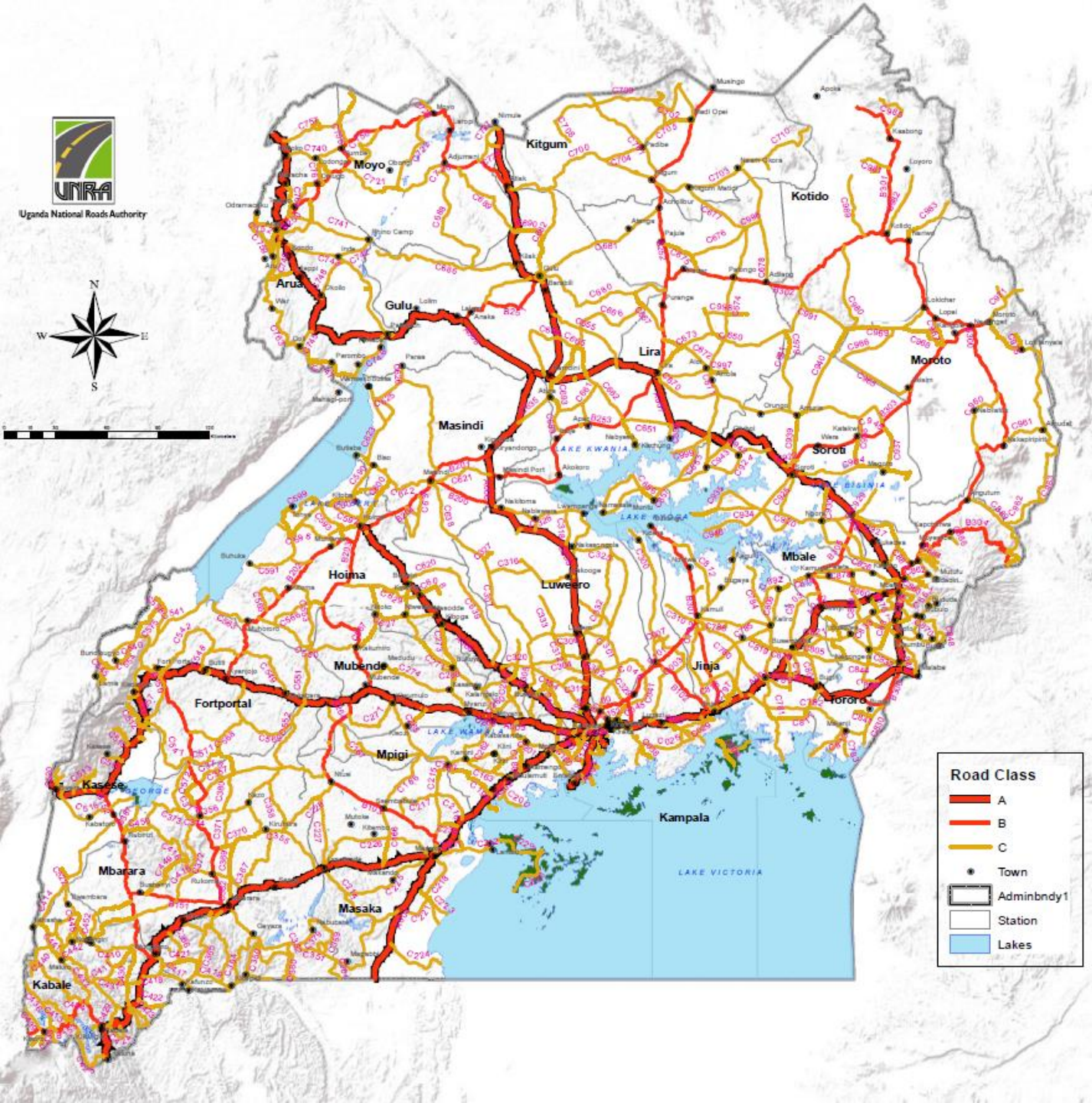
Network Classification





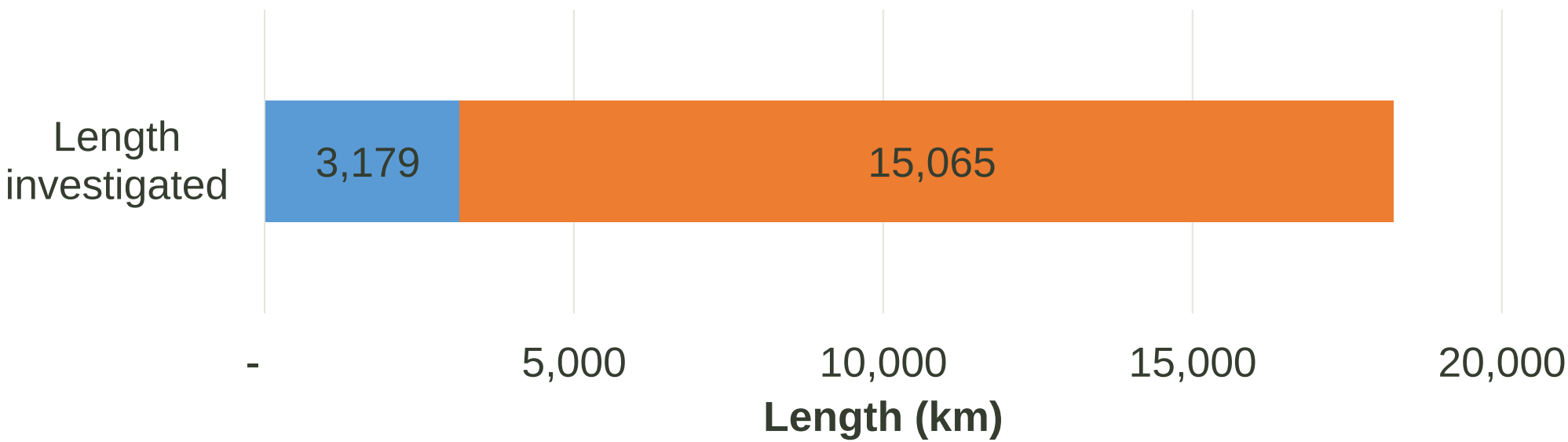
Network Classification

Functional Classification		Description
A	2594	International Trunk Roads
B	2843	National Trunk Roads
C	15108	Primary Roads
Total	20544	





Analyzed Network



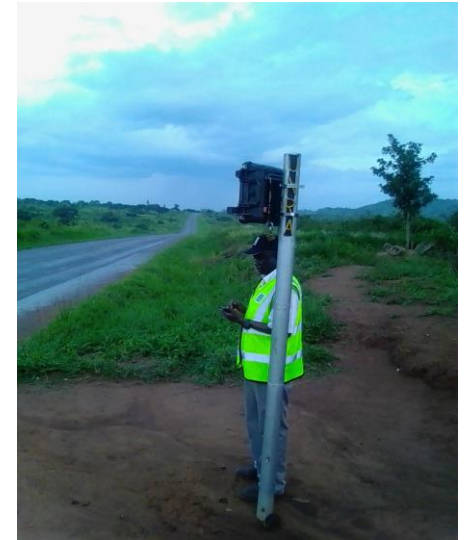
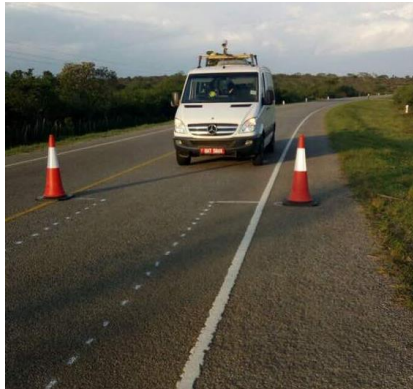
■ Paved ■ Unpaved

89% of network, 72% of the paved roads and 93% of the unpaved roads.



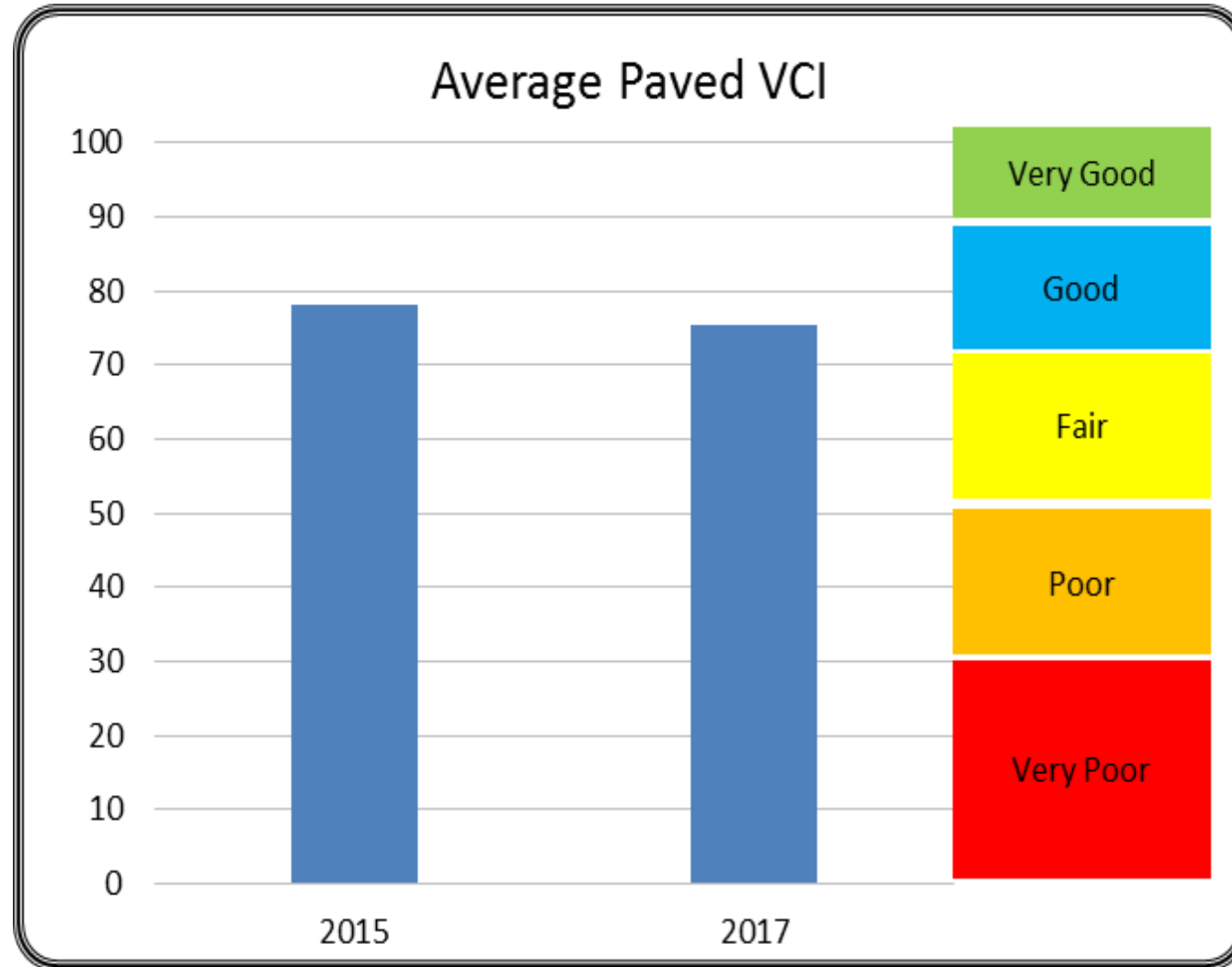
Data Collection Methods used

- Visual Condition Assessment - Mobicap
- Machine surface measurements - ROMDAS
- Automated/ Manual Traffic Collection
- Inventory Surveys
- Bridge Inventory and condition





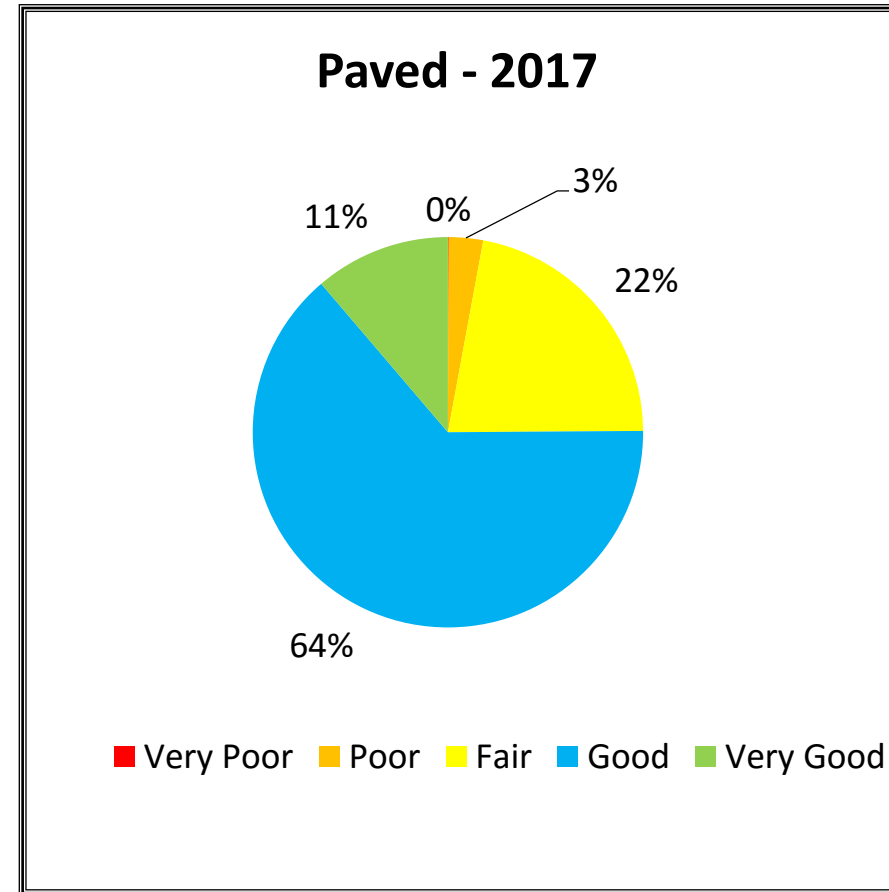
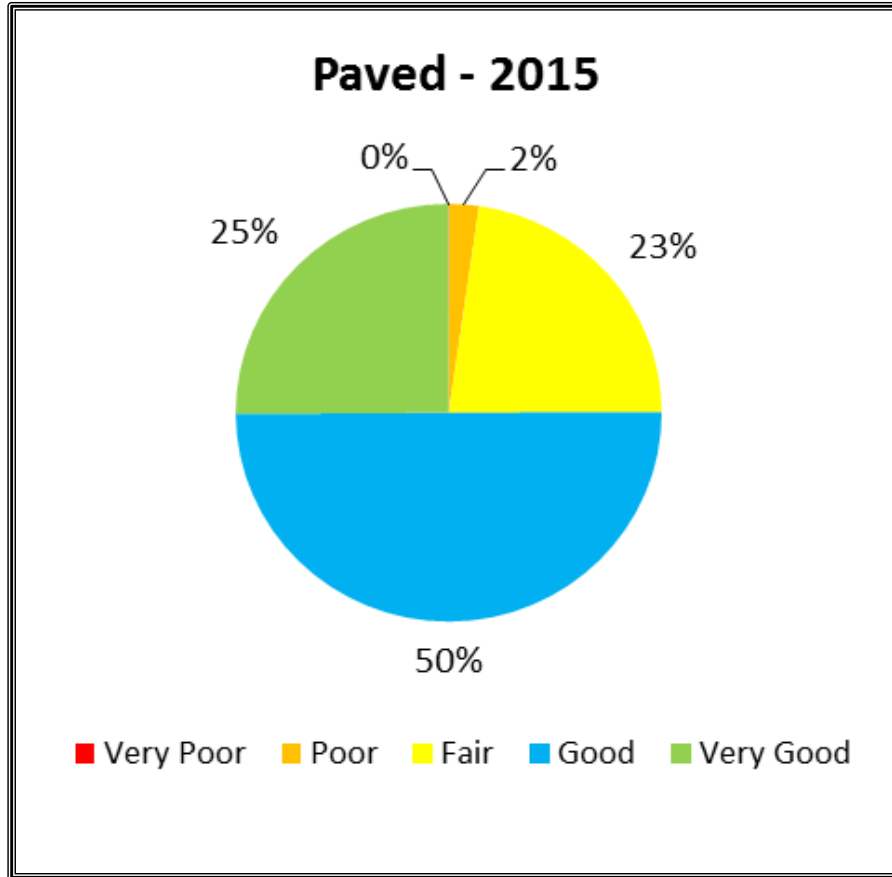
Road Condition – Paved Roads



VCI is 75 % -
2017.
3% Decline
from 2015



Road Condition – Paved Roads

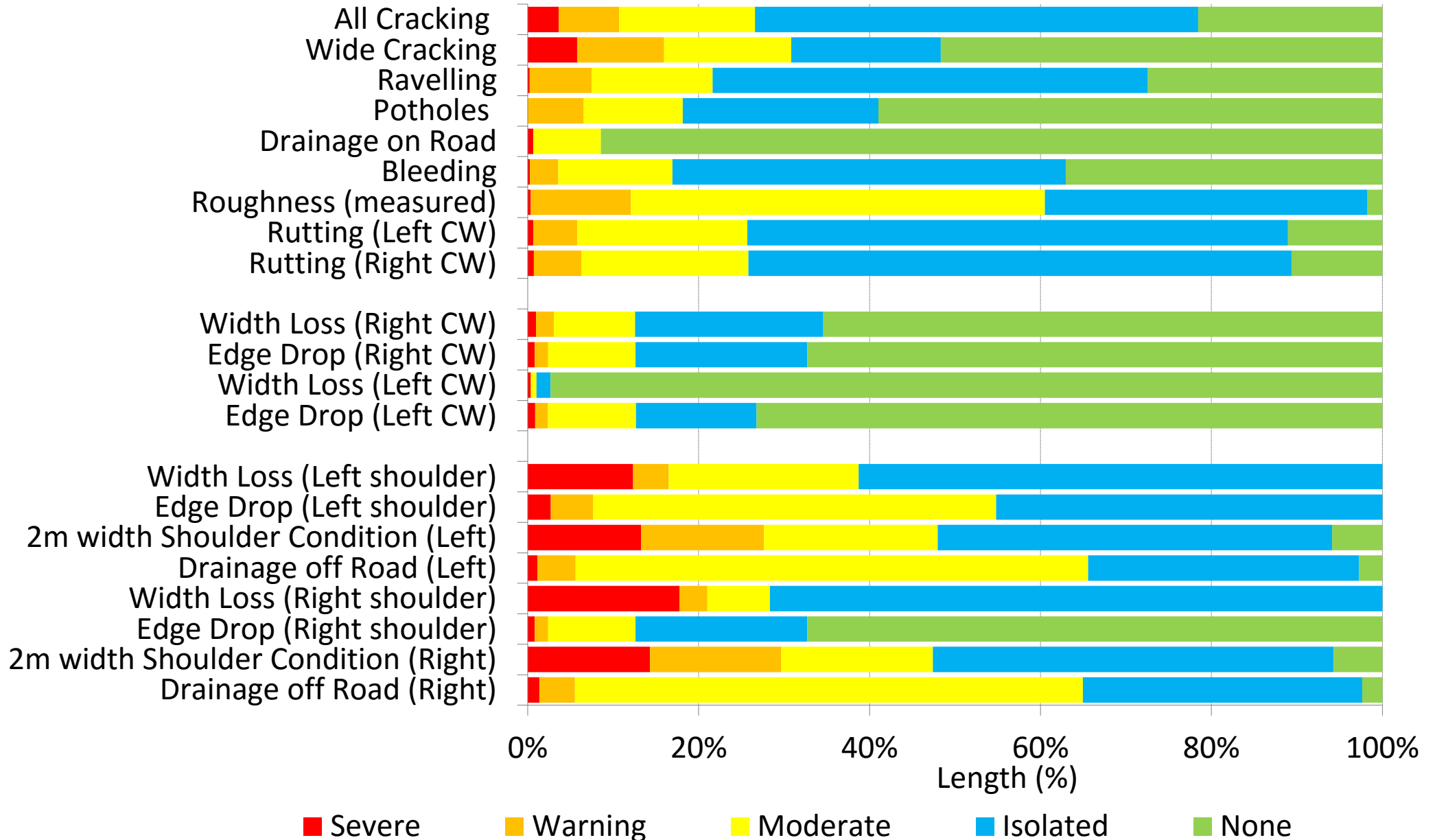




Road Condition – Paved Roads

- Overall, the paved road network is in good condition with less than 5% in poor and very poor condition, indicating a low backlog for rehabilitation.
- The 22% roads in fair condition indicate backlog for preventive maintenance and if not addressed, will be subject to MAD leading to very poor condition in a short time thus increasing the rehabilitation backlog in a very short time.

Paved Roads - 2017





Road Condition – Paved Roads



Busega - Masaka



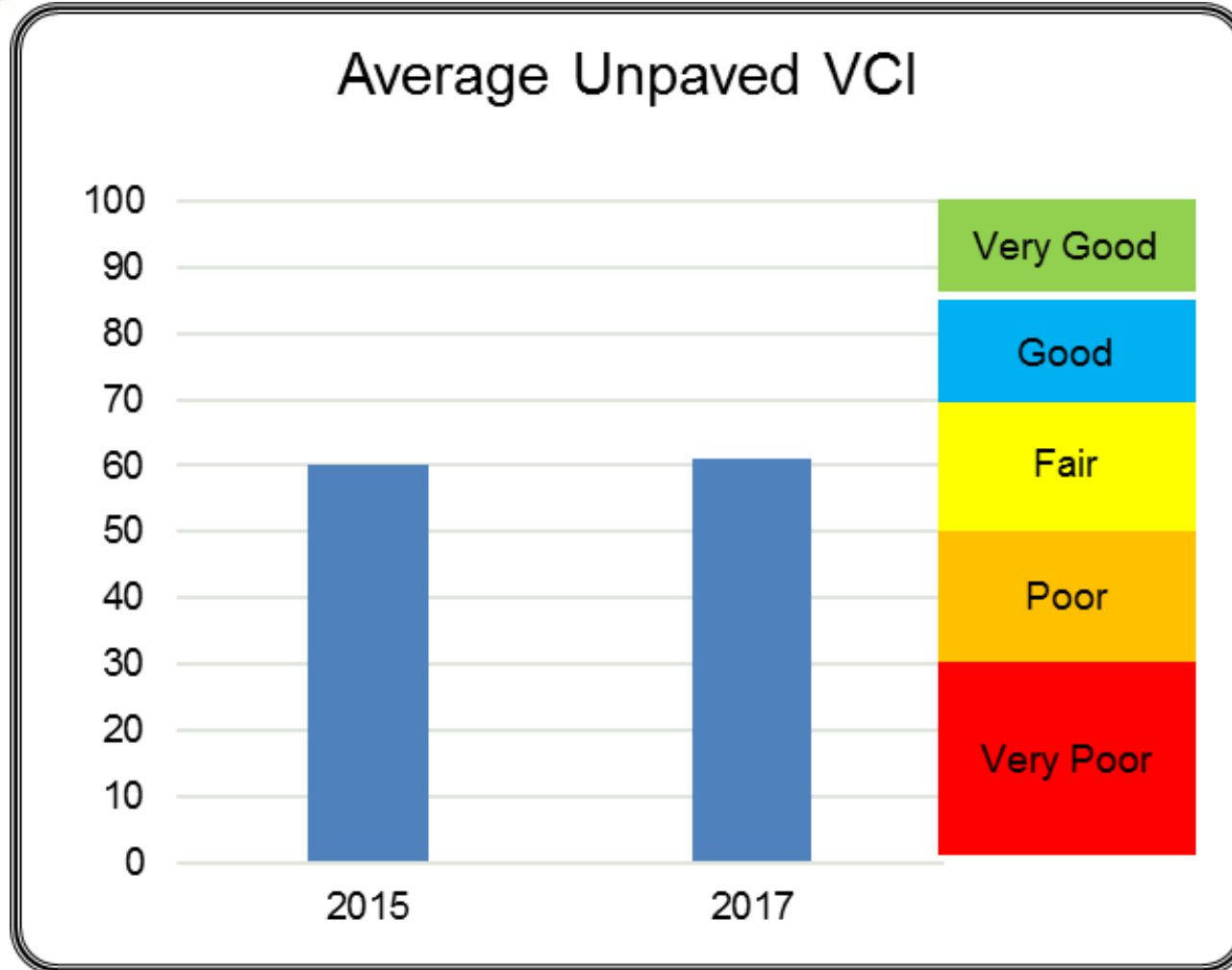
Mbarara - Kabale



Semuto - Kapeeka



Road Condition – Unpaved Roads

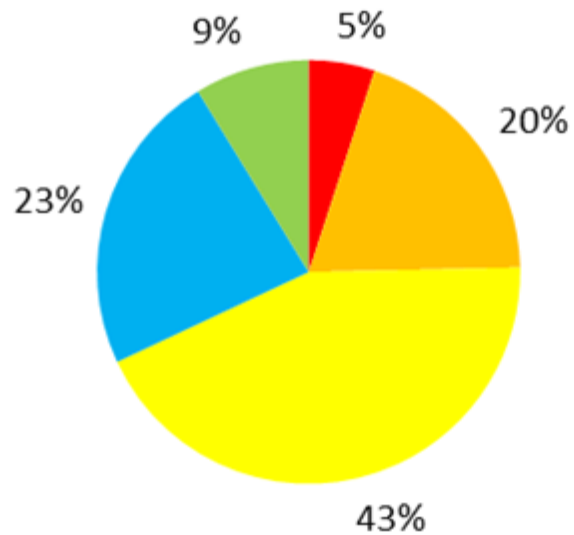


GCI is 62% -
2017.
1%
Improvement
from 2015



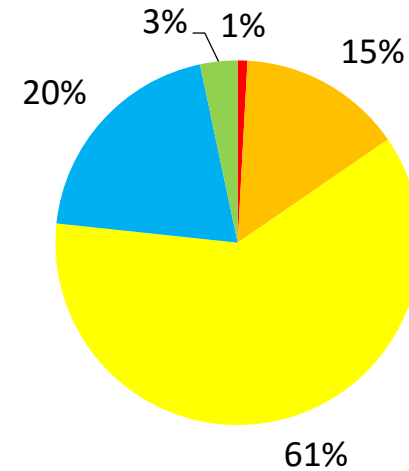
Road Condition – Unpaved Roads

Unpaved - 2015



Very Poor Poor Fair Good Very Good

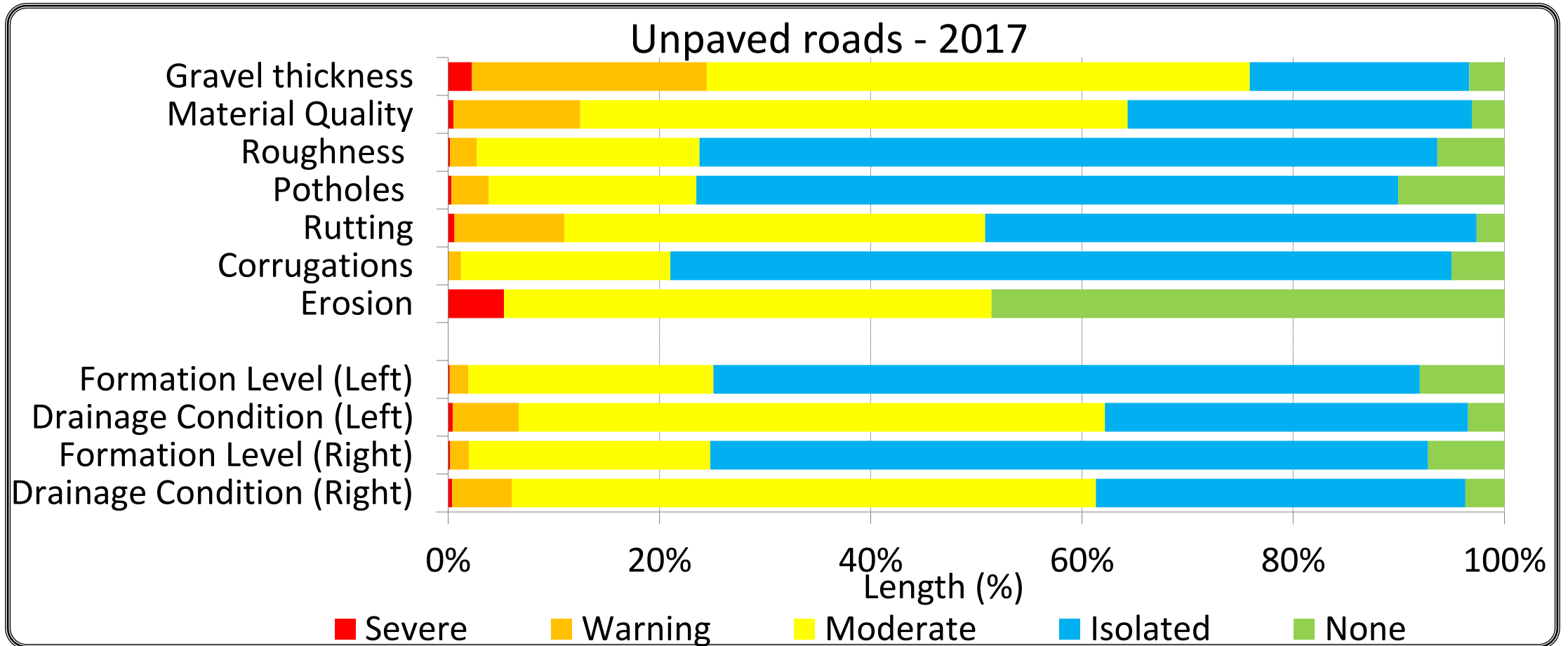
Unpaved - 2017



Very Poor Poor Fair Good Very Good

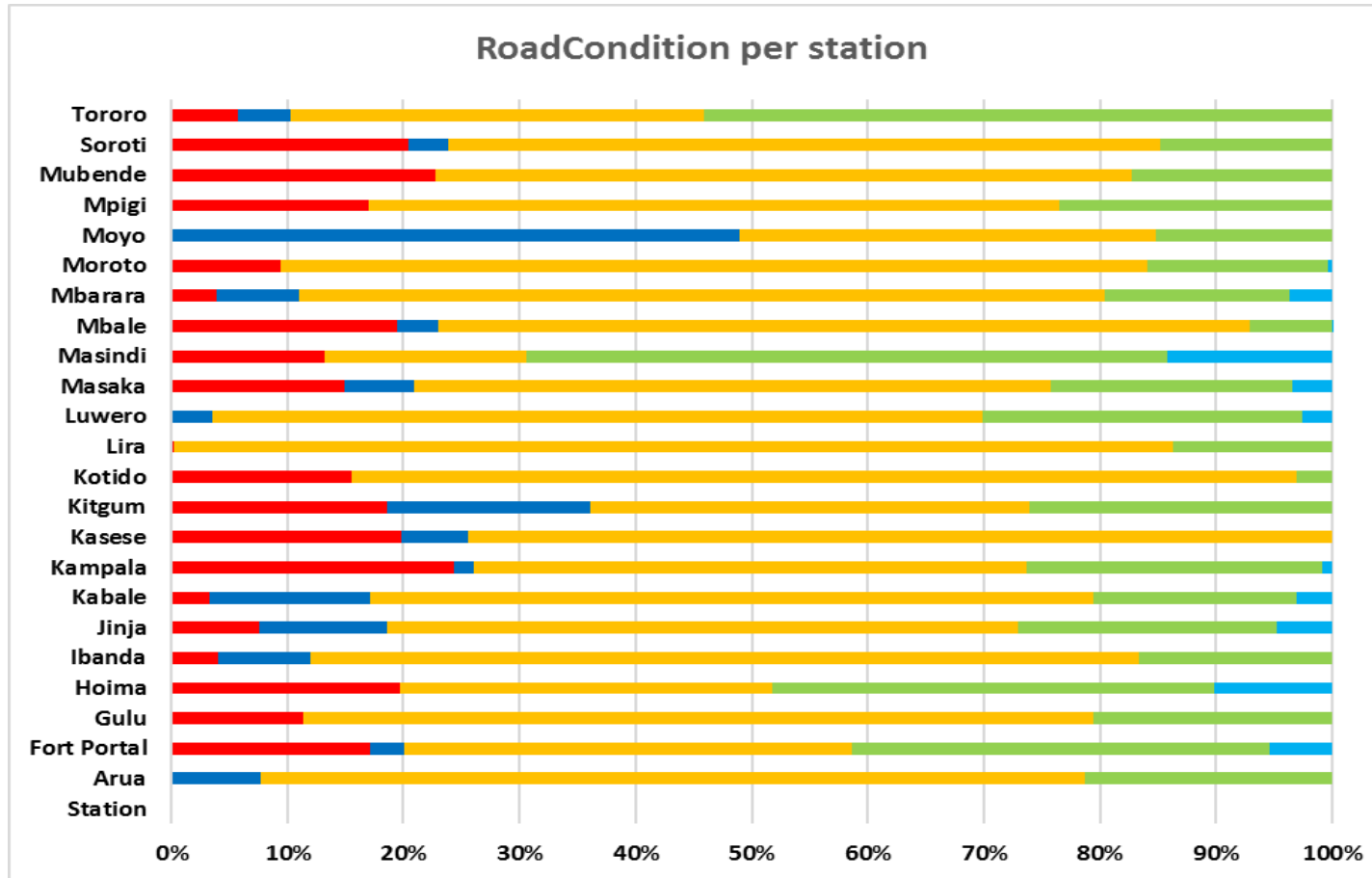


Road Condition – Unpaved Roads





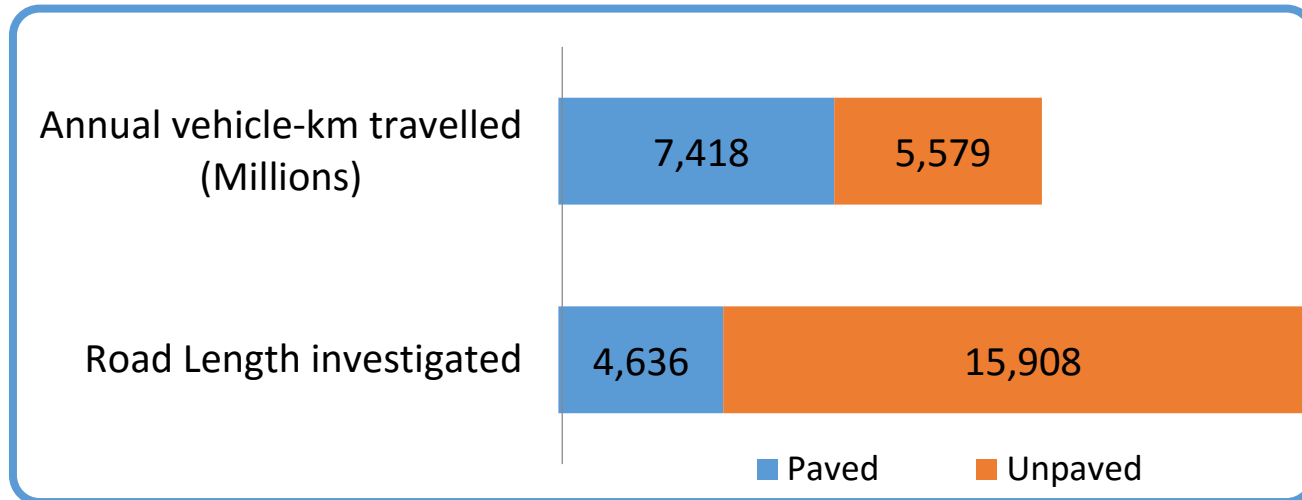
Summary Road Condition per station





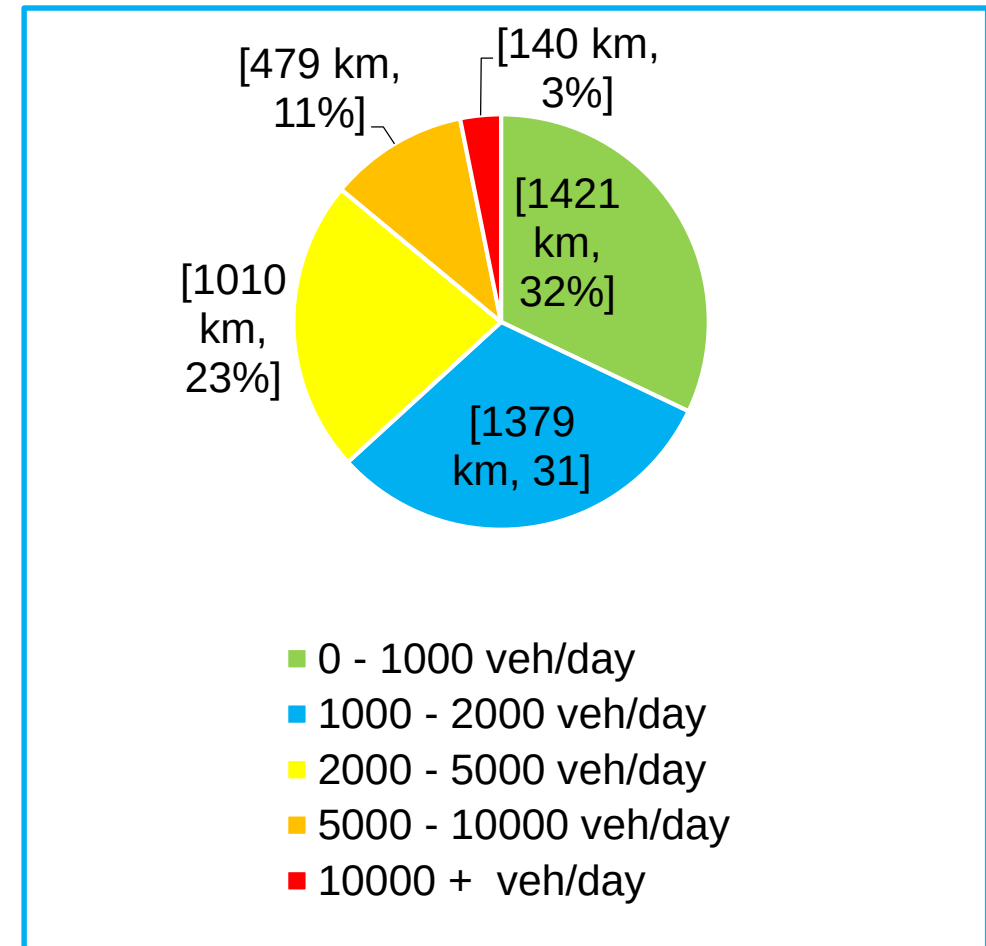
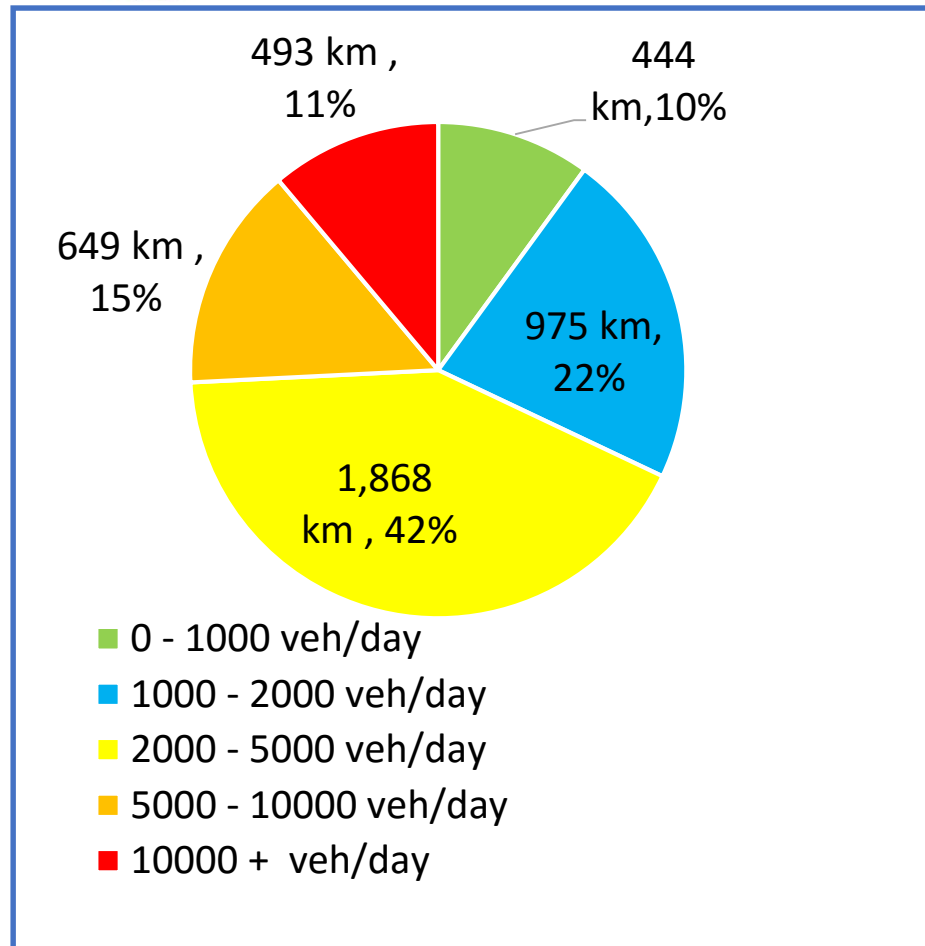
TRAFFIC

- 12,977 mil veh-km in total. (AADT by Length)
(7,418 mil veh-km paved roads and 5,579 mil veh-km unpaved roads)
- 25% paved network < 5,000 veh/day and 60% unpaved roads < 500 veh/day.
- Paved road network ~ 20% of road network carries about 60% of the network's traffic



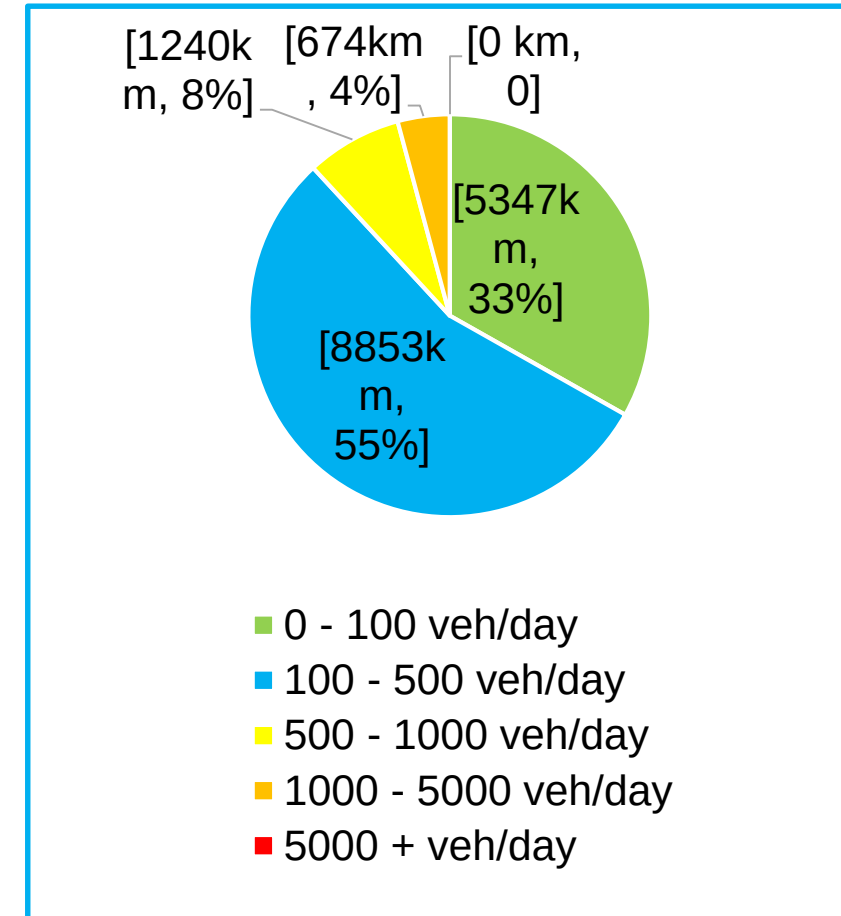
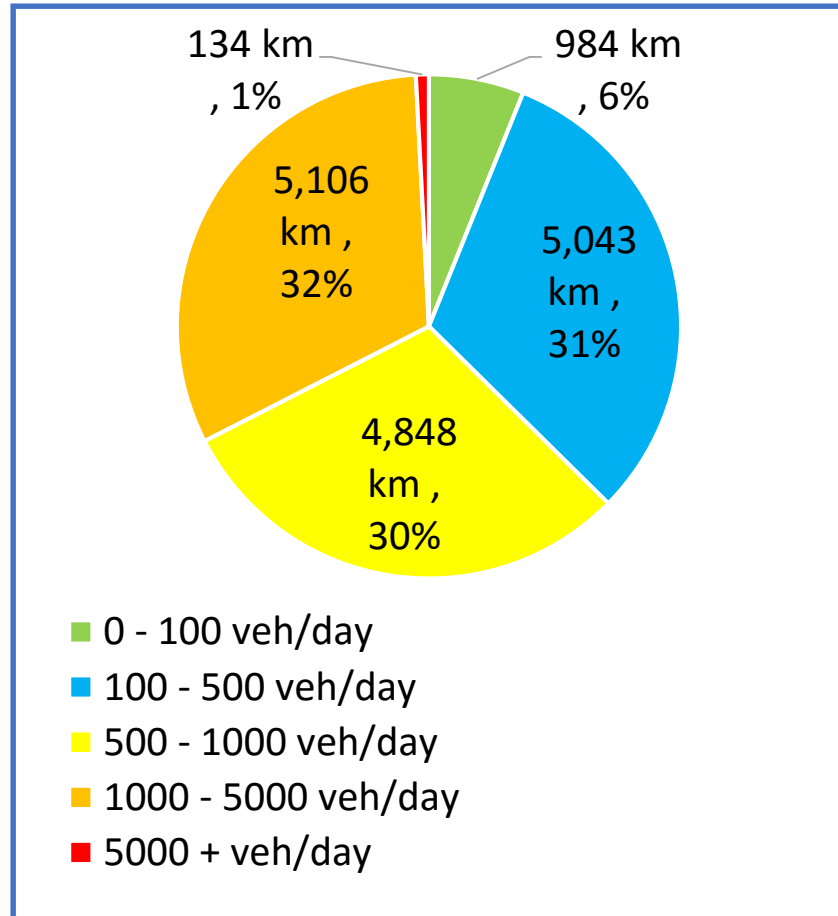


Traffic distribution Paved Roads





Traffic distribution Unpaved Roads





Boda bodas – Here to stay!!

Impacts

- Affect planning and design considerations
- Road usage (Other motorists)
- Fatal accidents thus Increased health budget

Reasons

- Easy maneuver through congested roads thus saving time
- Poor Road condition
- Convenience (Fewer stops) and door to door

Way forward

- Improve policy
- Improve condition
- Make them safe
- Incorporate them in the designs and planning (AADT)



Way forward on Boda bodas

- Traffic beyond 200-300 veh/day means potential for upgrading.
Unpaved roads - 63% >500 incl boadsand without

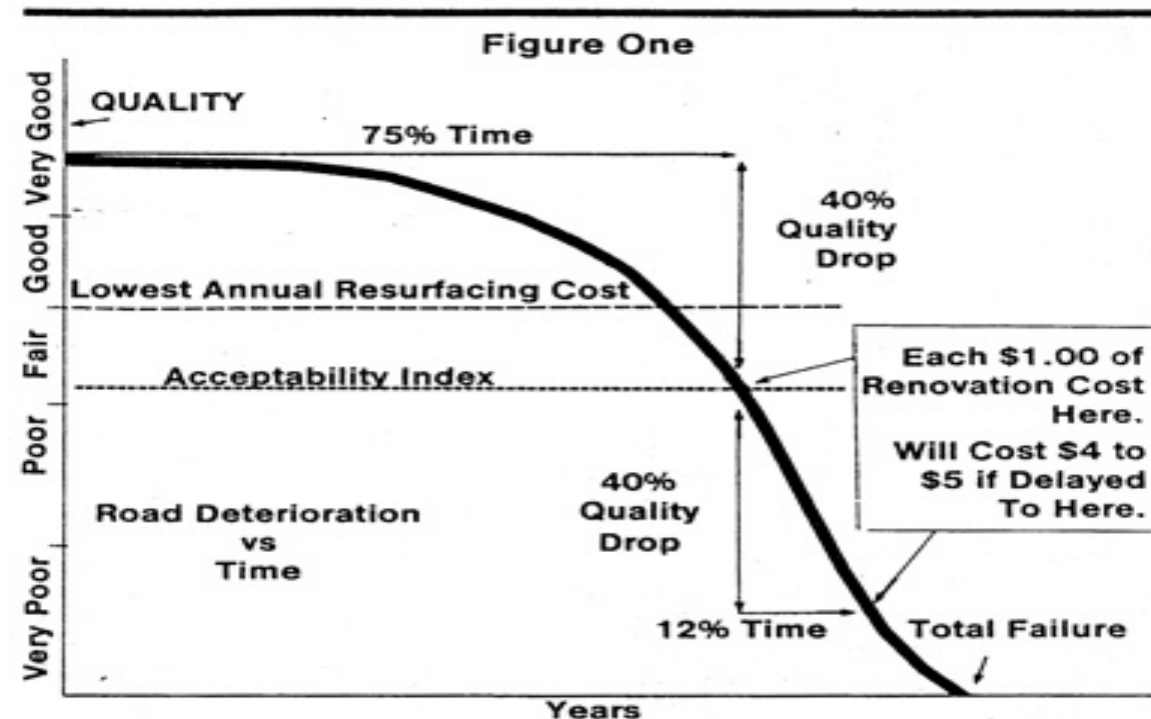
How do we deal with them!

- Exclude them during network level analysis for treatment triggers OR convert them to standard vehicles
- Include them during design and capacity analysis and construct designated lanes for them
- *Chase them off the roads*

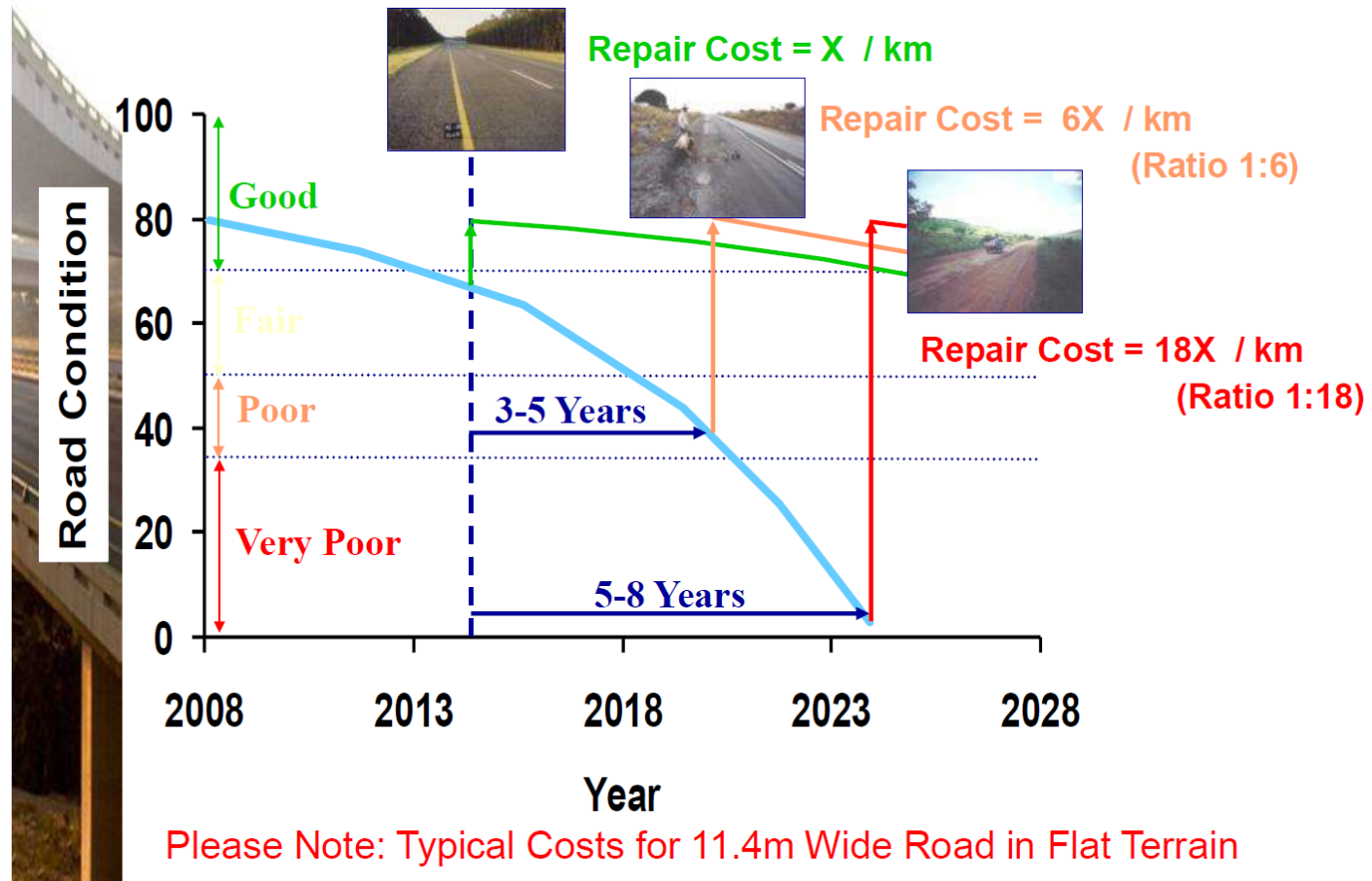


NETWORK NEEDS

- Performance prediction models model the future performance of the roads with an objective of the optimisation through LCCA to preserve the network condition.



The price we pay





NEEDS ANALYSIS Inputs

- Present condition;
- Pavement performance models;
- Maintenance, regravelling and upgrading treatments and their costs;
- Vehicle operating cost; and
- Optimisation function

LIFE CYCLE COST ANALYSIS

YEAR SMILES

Input

Data

Network definition



Current and Historic Condition data

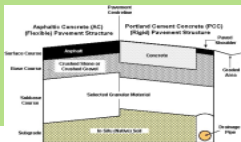
- Visual survey
- Profile data



Traffic data



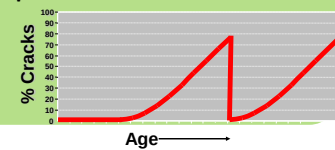
Pavement data



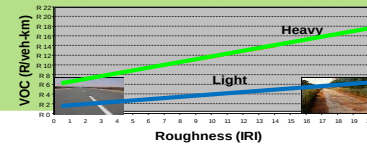
Models & Information

Pavement performance models

- HDM-4



Vehicle operating cost models



Maintenance

- Committed treatments
- Treatments
- Triggers
- Unit costs
- Policies



Budget Scenarios

- MTEF
- Any other fund level



Life cycle cost analysis with dTIMS



Output

Decision support

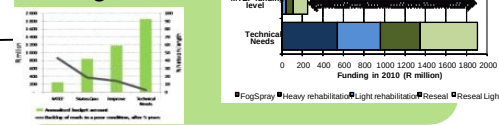
Efficiency frontier

- Generate strategies
- B vs C

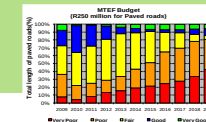


Support to planning and budgeting

Funding need



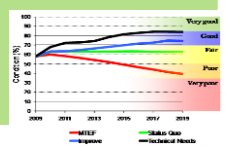
Determine effect of maintenance policies on road network



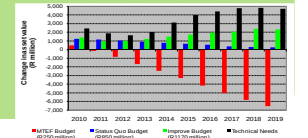
Reporting

Condition consequences

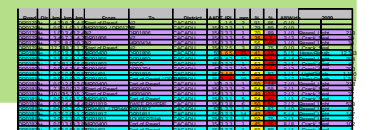
- Predicted CI
- Predicted VOC
- Predicted EUC
- Predicted Asset Value



Needs Report



Maintenance, Rehabilitation and Upgrading Plans



Geographic maps

- MTEF
- Any other fund level



TREATMENT STRATEGIES

The funding scenarios discussed and analysed concern the budget values for the following treatments only:

- Preventive maintenance (light reseal/reseal) of paved roads.
- Rehabilitation of paved roads (light / heavy).
- Regravelling of unpaved roads.
- Upgrading of unpaved roads to paved standards.



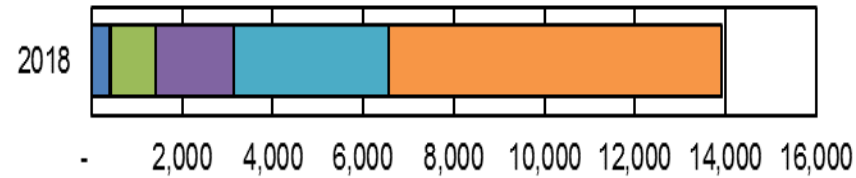
TREATMENT COSTS

- Upgrade to standard paved – **950,000 USD/Km**, traffic > **200 vpd**, savings in RUC > upgrade costs, positive TTC
- Regravel – **13,500USD/Km**, Gravel thickness < 30mm
- Heavy Rehabilitation – **110USD/M2**, Condition poor, IRI > 6.6
- Light Rehabilitation – **35USD/M2**, condition fair, IRI < 6.6
- Reseal – **10USD/Km**, AADT > 4000, IRI < 4.2
- Reseal Light – **8USD/Km**, AADT < 4000, IRI < 4.2



Immediate Technical Needs

- Remove all backlog

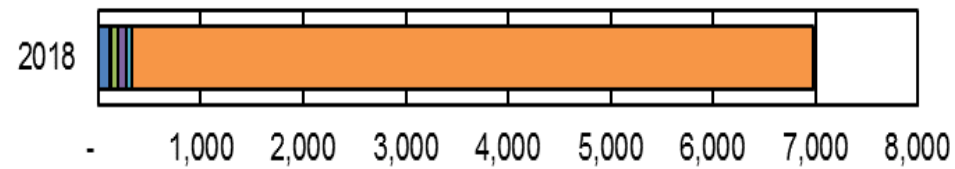


	2018
■ Heavy Rehabilitation	384
■ Light Rehabilitation	46
■ Reseal	988
■ Light Reseal	1,722
■ Regravel	3,435
■ Upgrade to Paved	7,330

Annual length of road to be treated
(km)



What it costs



	2018
■ Heavy Rehabilitation	114
■ Light Rehabilitation	12
■ Reseal	66
■ Light Reseal	88
■ Regravel	46
■ Upgrade to Paved	6,657

Annual budget allocated towards maintenance and rehabilitation
(\$ Million)

Optimization

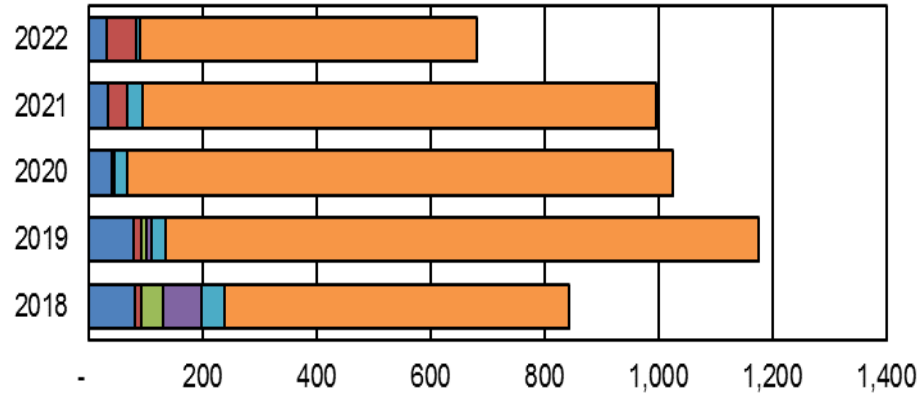
Funding Scenarios Investigated



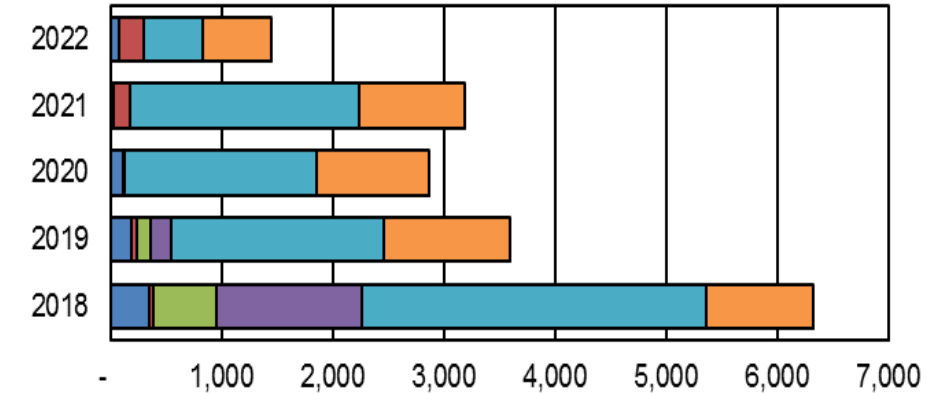
Year	2018/19	2019/20	2020/21	2021/22	2022/23
Total Budget Received UGX	3,130	4,358	3,811	3,692	2,528
Total Budget Received USD	843,777,358	1,174,688,140	1,027,195,957	995,260,647	681,344,474
Total Budget Requested UGX	4,783	4,783	4,783	4,783	4,783
Total Budget Requested USD	1,289,329,919	1,289,329,919	1,289,329,919	1,289,329,919	1,289,329,919



Received budget (US\$ 944 Annually)



Annual budget allocated towards maintenance and rehabilitation
(\$ Million)



Annual length of road to be treated
(km)



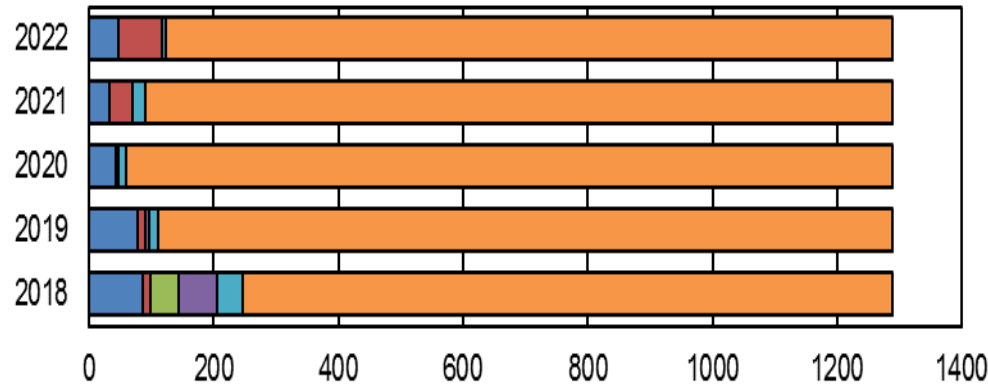
Received budget

On average 10% (**US\$ 100 mil**) should be allocated annually to paved roads over this period.

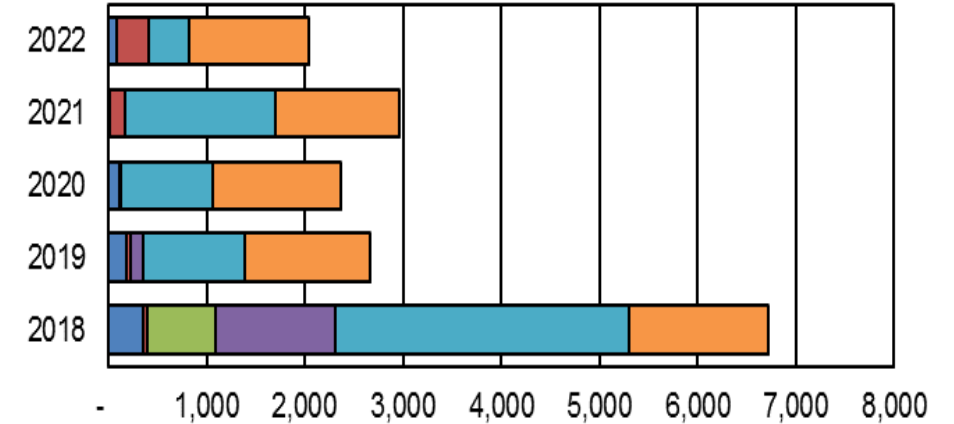
- **US\$ 76 mil** of it be invested to rehabilitation of roads and **US\$ 24 mil** to resealing.
- The result would be **1,204 km** rehabilitated and **2,189 km** resealed.
- On average 89% (**US\$ 844 mil**) should be allocated annually to unpaved roads.
- **US\$ 818 mil** for upgrading of unpaved roads and **US\$ 25 mil** to regravelling.
- The result would be **4,679 km** upgraded and **9,340 km** regravelled.



Requested Budget



Annual budget allocated towards maintenance and rehabilitation
(\$ Million)



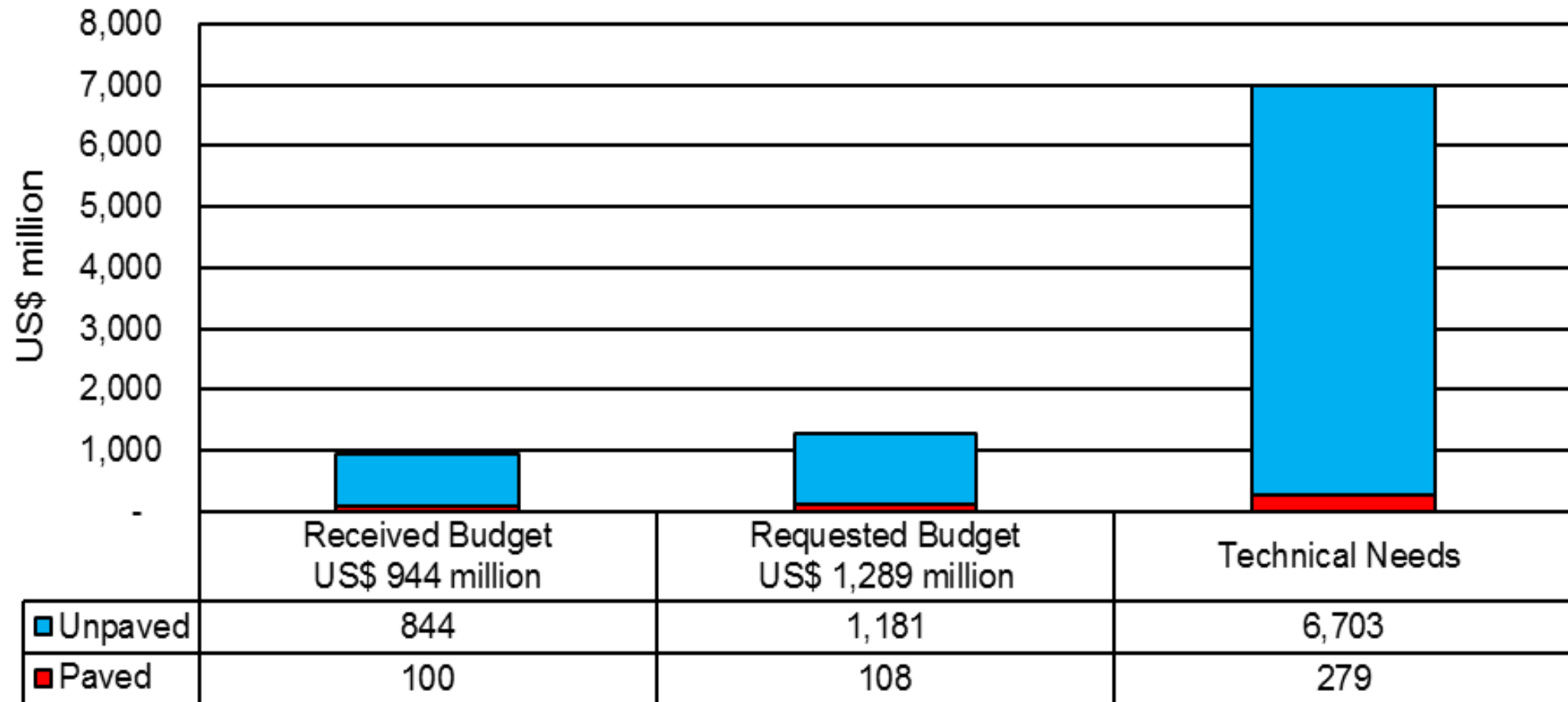
Annual length of road to be treated
(km)



Requested Budget

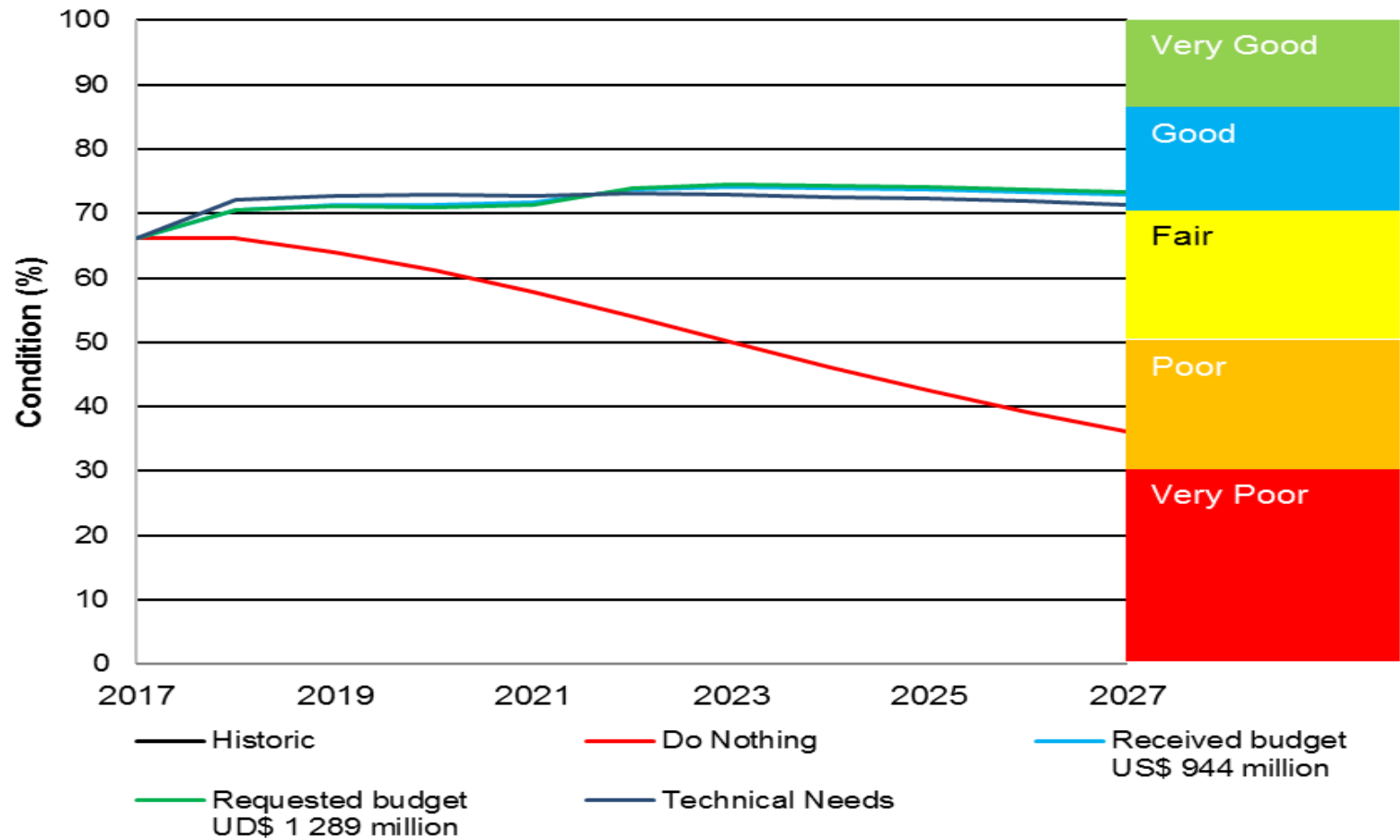
- On average, **8% (US\$ 108 mil)** be allocated annually to paved roads
- **US\$ 85 mil** of it be invested towards rehabilitation and the remaining **US\$ 23 mil** to resealing.
- Over the five year period **1,334 km** of road could be rehabilitated and **2,045 km** could be resealed.
- **91% (US\$ 1,181 mil)** be allocated to unpaved roads annually.
- **US\$ 1,162 million** annually to upgrading of unpaved roads and the remaining **US\$ 19 mil** towards regravelling.
- In total over the five year period **6,490 km** of road could be upgraded and **6,892 km** could be regravelled.

Comparison of Different Funding Levels



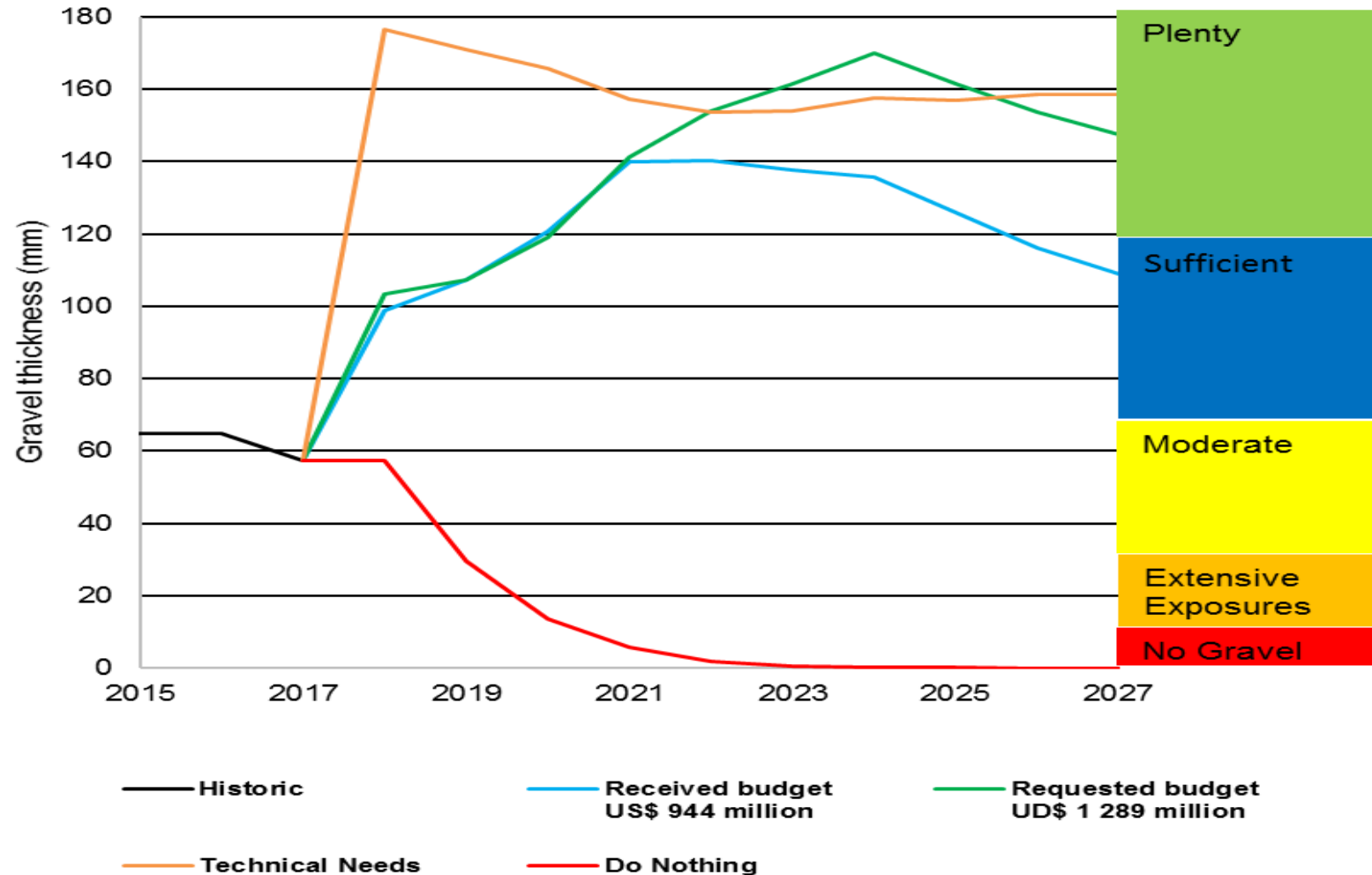


Predicated Condition of paved Roads





Predicated Condition of Unpaved Roads





Do nothing scenario

- We run it for comparison of strategies
- Used as a baseline for calculating the triggers
- Could happen if government changes strategies
- And In extreme cases of war (Syria, Libya, Iraq)



Vehicle Operating Costs

- The current VOC is **US\$ 1,278** per 1000 veh- km travelled.
- VOC by 2026 for:
- Do Nothing scenario is **US\$ 1,283**
 - Received Budget is **US\$ 835**
 - Requested Budget is **US\$ 739**

Generated by RED and used to calculate the economic benefit and economic consequence of maintenance strategy and budget constraints

2015/16 VOC US\$ 1,040 per 1000 veh- km travelled.

VOC by 2025 for: Do Nothing scenario was US\$ 1,126.

Received Budget was US\$ 800

Requested Budget was US\$ 725



Excess User Costs

- The current EUC is **US\$ 1,191** per 1000 vehicle-kilometres travelled.
- EUC by 2026 for:
- Do Nothing scenario is **US\$ 1,893**
 - Received Budget is **US\$ 260**
 - Requested Budget is **US\$ 234**



Asset Values -Paved Roads

- The Current Depreciated Replacement Cost (CDRC) for the paved road network is **US\$ 2.93 billion** (Decline from **US\$3.2 billion** in 2015) while the Current Replacement Cost (CRC) is **US\$ 4.33 billion** (Increment from **US\$4.0 billion** in 2015).
- The CDRC is **68%** of the CRC, equivalent to asset loss of **32%**. A CDRC to CRC ratio greater than 80% is indicative of a well-maintained road network and therefore the current performance of the paved road network has declined from 79%.



Asset Values -Unpaved Roads

- For the unpaved network the CDRC is **US\$ 649 million** and the CRC is **US\$ 1,699 million**
- The CDRC is currently 38 per cent of the CRC.



Assumptions

- The unit cost per km includes all costs to construct a road
- There will be no excessive delayed payments and thus no unplanned interest paid to service providers
- Inflation will be fairly stable as predicted
- Traffic growth is uniform all through at 6%
- The discount rate is 8%
- Supporting processes are efficient



Assumptions

- All construction processes are undertaken in one year for the delivery of those triggered km
- Projects “committed” but not certain of funding thus not started yet have been excluded
- Traffic assignment is accurate
- That land is available



Department Enhancements

- Exposure to mature asset Management institutions
- Upgrade to latest systems. Laser, 360 cameras

